BCA-II SEMESTER

MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE (BCA-108)

Unit 1:

SYLLABUS :

Basic Statistics: Measure of Central Tendency, Preparing Frequency Distribution Table, Mean, Median, Mode, Measure of Dispersion: Range, Variance and Standard Deviations, Correlation and Regression.

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Introduction

In the modern world of computers and information technology, the importance of statistics is very well recognized by all the disciplines. Statistics has originated as a science of statehood and found applications slowly and steadily Agriculture, Economics, Commerce, Biology, Medicine, Industry, planning, education and so on. As on date there is no other human walk of life, where statistics cannot be applied.

Meaning of Statistics:

Statistics has been defined differently by different authors and each author has assigned new limits to the field which should be included in its scope.

A.L. Bowley defines, "Statistics may be called the science of counting". At another place he defines, "Statistics may be called the science of averages". Both these definitions are narrow and throw light only on one aspect of Statistics.

According to **King,** "The science of statistics is the method of judging collective, natural or social, phenomenon from the results obtained from the analysis or enumeration or collection of estimates".

According to **Wallis and Roberts** "Statistics is a body of methods for making wise decisions on the face of uncertainty."

According to **Edward N. Dubois** "Statistics is a body of methods for obtaining and analyzing numerical data in order to make better decisions in an uncertain world."

In the words of **Croxton & Cowden**, "Statistics may be defined as the collection, presentation, analysis and interpretation of numerical data".

Horace Secrist has given an exhaustive definition of the term statistics in the plural sense. According to him: "By statistics we mean aggregates of facts affected to a marked extent by a multiplicity of causes numerically expressed, enumerated or estimated according to reasonable standards of accuracy collected in a systematic manner for a pre-determined purpose and placed in relation to each other".

Scope of Statistics:

Statistics is not a mere device for collecting numerical data, but as a means of developing sound techniques for their handling, analyzing and drawing valid inferences from them. Statistics is applied in every sphere of human activity– social as well as physical – like Biology, Commerce, Education, Planning, Business Management, Information Technology, etc. It isalmost impossible to find a single department of human activity where statistics cannot be applied. We now discuss briefly the applications of statistics in other disciplines.

1. **Statistics in Industry:** In industrial line, statistics plays an important role such as in quality control and production engineering various control charts are used to maintain a certain quality level and different inspection plans are used in production engineering. Even to find average life of some products such as electric bulb, sampling technique is used.

2. **Statistics in Commerce:** In present times, there is a very tough competition in almost every business. Also fashions, likings/tastes, requirements, trends, levels of qualities, technologies, etc. are changing very fast. So for the success of the business, it has become necessary for a business man to know the coming trend of market in advance or as soon as possible. This can also be achieved only with the help of market survey, which requires statistical techniques.

3. **Statistics in Agriculture:** Presently there are a number of varieties of seeds for a particular crop. Also different types of fertilizers are available in the market. For a good yield, it has become necessary to know that which one is better. This job is again done by a very popular and widely used test known as Analysis of variance (ANOVA) discovered by Professor R.A. Fisher.

4. **Statistics in Education:** Statistics is widely used in education. Research has become a common feature in all branches of activities. Statistics is necessary for the formulation of policies to start new course, consideration of facilities available for new courses etc. There are many people engaged in research work to test the past knowledge and evolve new knowledge. These are possible only through statistics.

5. **Statistics in Planning:** Every institution/organization plans for its future targets. Now, a days for a good planning, it has became necessary to analysis the statistical data according to the field of interest such as availability of raw material, consumption, investment, resources available, income, expenditure, quality needed, etc. In order to analysis these types of data, one has to totally depend on the statistical techniques. Thus statistics is essential for planning.

6. **Statistics in Medicine:** Statistics also plays an important role in the field of medicine. The hypothesis of the types:

a) Drug A is better than drug B.

b) Smoking and cancer are associated.

c) Smoking and TB are associated.

All are tested using t-test or Chi2-test as the case may be. Statistics also find its application in clinical trials.

7. **Statistics and Insurance Sector:** Whole insurance sector totally depends on the statistical data and different concepts of probability theory. Life tables lies in the heart of human

insurances. Curate future life time and complete future life time, of a life are calculated using concept of random variables and their expected values. Due to the large use of statistics in insurance sector, a new branch of statistics known as Actuarial statistics has been started in some institutes throughout the world.

Statistics in Modern applications: Recent developments in the fields of computer technology and information technology have enabled statistics to integrate their models and thus make statistics a part of decision making procedures of many organizations. There are so many software packages available for solving design of experiments, forecasting simulation problem etc. SYSTAT, a software package offers mere scientific and technical graphing options than any other desktop statistics package.

Limitations of statistics:

Statistics with all its wide application in every sphere of human activity has its own limitations. Some of them are given below.

1. **Statistics is not suitable to the study of qualitative phenomenon:** Since statistics is basically a science and deals with a set of numerical data, it is applicable to the study of only these subjects of enquiry, which can be expressed in terms of quantitative measurements. As a matter of fact, qualitative phenomenon like honesty, poverty, beauty, colour, intelligence etc, cannot be expressed numerically and any statistical analysis cannot be directly applied on these qualitative phenomenon's. Nevertheless, statistical techniques may be applied indirectly by first reducing the qualitative expressions to accurate quantitative terms. For example, the intelligence of a group of students can be studied on the basis of their marks in a particular examination.

2. **Statistics does not study individuals:** Statistics does not give any specific importance to the individual items; in fact it deals with an aggregate of objects. Individual items, when they are taken individually do not constitute any statistical data and do not serve any purpose for any statistical enquiry.

3. **Statistical laws are not exact:** It is well known that mathematical and physical sciences are exact. But statistical laws are not exact and statistical laws are only approximations. Statistical conclusions are not universally true. They are true only on an average.

4. **Statistics table may be misused:** Statistics must be used only by experts; otherwise, statistical methods are the most dangerous tools on the hands of the inexpert. The use of statistical tools by the inexperienced and untraced persons might lead to wrong conclusions. Statistics can be easily misused by quoting wrong figures of data. As King says aptly 'statistics are like clay of which one can make a God or Devil as one pleases'.

5. **Statistics is only, one of the methods of studying a problem:** Statistical methods do not provide complete solution of the problems because problems are to be studied taking the background of the countries culture, philosophy or religion into consideration. Thus the statistical study should be supplemented by other evidences.

Preparing Frequency Distribution Table

Q. What is a frequency distribution?

The frequency of a value is the number of times it occurs in a dataset. A frequency distribution is the pattern of frequencies of a variable. It's the number of times each possible value of a variable occurs in a dataset.

Types of frequency distributions

There are four types of frequency distributions:

1. **Ungrouped (Discrete) frequency distributions:** The number of observations of each value of a variable. You can use this type of frequency distribution for categorical variables.
2. **Grouped (Continuous) frequency distributions:** The number of observations of each class interval of a variable. Class intervals are ordered groupings of a variable's values. You can use this type of frequency distribution for quantitative variables.
3. **Relative frequency distributions:** The proportion of observations of each value or class interval of a variable. You can use this type of frequency distribution for any type of variable when you're more interested in comparing frequencies than the actual number of observations.
4. **Cumulative frequency distributions:** The sum of the frequencies less than or equal to each value or class interval of a variable. You can use this type of frequency distribution for ordinal or quantitative variables when you want to understand how often observations fall below certain values.

Ungrouped (Discrete) frequency distributions:

The discrete frequency distribution shows the frequency of an item in each individual data value rather than groups of data values.

Q. How to make an ungrouped frequency table?

Step 1: Create a table with two columns and as many rows as there are values of the variable. Label the first column using the variable name and label the second column "Frequency." Enter the values in the first column.

Step 2: Count the frequencies. The frequencies are the number of times each value occurs. Enter the frequencies in the second column of the table beside their corresponding values.

More especially if your dataset is large, it may help to count the frequencies by tallying. Add a third column called "Tally." As you read the observations, make a tick mark in the appropriate row of the tally column for each observation. Count the tally marks to determine the frequency.

Example: In a survey of 40 families in a village, the number of children per family was recorded and the following data obtained.

1	0	3	2	1	5	6	2	2	1
0	3	4	2	1	6	3	2	1	5
3	3	2	4	2	2	3	0	2	1
4	5	3	3	4	4	1	2	4	5

Represent the data in the form of a discrete frequency distribution.

Solution: Frequency distribution of the number of children

<table>
<tr><th>No. of Children</th><th>Tally Marks</th><th>Frequency</th></tr>
<tr><td>0</td><td>|||</td><td>3</td></tr>
<tr><td>1</td><td>卌 ||</td><td>7</td></tr>
<tr><td>2</td><td>卌 卌</td><td>10</td></tr>
<tr><td>3</td><td>卌 |||</td><td>8</td></tr>
<tr><td>4</td><td>|||||</td><td>6</td></tr>
<tr><td>5</td><td>||||</td><td>4</td></tr>
<tr><td>6</td><td>||</td><td>2</td></tr>
<tr><td></td><td>Total</td><td>40</td></tr>
</table>

Grouped (Continuous) frequency distributions:

A continuous frequency distribution is a series in which the data are classified into different class intervals without gaps and their respective frequencies are assigned as per the class intervals and class width.

Before we begin to construct continuous frequency distribution table, there are some basic technical terms when a continuous frequency distribution is formed or data are classified according to class intervals.

a) Class Limits:

The class limits are the lowest and the highest values that can be included in the class.
For example, take the class 30-40. The lowest value of the class is 30 and highest class is 40. The two boundaries of class are known as the lower limits and the upper limit of the class respectively. The lower limit of a class is the value below which there can be no item in the class. The upper limit of a class is the value above which there can be no item to that class.

If the class 60-79, 60 is the lower limit and 79 is the upper limit, i.e. in the case there can be no value which is less than 60 or more than 79. The way in which class limits are stated depends upon the nature of the data. In statistical calculations, lower class limit is denoted by L and upper class limit by U.

b) Class Interval:

The class interval may be defined as the size of each grouping of data. For example, 50-75, 75-100, 100-125… are class intervals. Each grouping begins with the lower limit of a class interval and ends at the lower limit of the next succeeding class interval.

c) Width or Size of the Class Interval:

The difference between the lower and upper class limits is called Width or size of class interval and is denoted by ‘C’.

For example: Consider class interval 300 – 400, then width of class interval is

$$C = U - L = 400 - 300 = 100.$$

d) Range:

The difference between largest and smallest value of the observation is called the Range and is denoted by ‘R’ i.e.

$$R = \text{Largest value} - \text{Smallest value} = L - S$$

e) Mid-Value or Mid-Point:

The central point of a class interval is called the mid value or mid-point. It is found out by adding the upper and lower limits of a class and dividing the sum by 2.

$$\text{Mid-value} = (L+U)/2$$

For example, if the class interval is 20-30 then the mid-value is (20+30)/2=25

Q. How to make a grouped (continuous) frequency table?

Step 1: Divide the variable into class intervals.

Step 2: Create a table with two columns and as many rows as there are class intervals. Label the first column using the variable name and label the second column “Frequency.” Enter the class intervals in the first column.

Step 3: Count the frequencies. The frequencies are the number of observations in each class interval. You can count by tallying if you find it helpful. Enter the frequencies in the second column of the table beside their corresponding class intervals.

Example: Wage distribution of 100 employees in form of continuous frequency distribution table.

Weekly Wages (Rs)	100-200	200-300	300-400	400-500	500-600
No. of Employees	8	16	22	38	16

Types of Class Intervals:

There are three methods of classifying the data according to class intervals namely

a) Exclusive method
b) Inclusive method
c) Open-end classes

a) Exclusive method:

When the class intervals are so fixed that the upper limit of one class is the lower limit of the next class; it is known as the exclusive method of classification. The following data are classified on this basis.

Expenditure (Rs.)	No. of Families
0-5000	60
5000-10000	95
10000-15000	122
15000-20000	83
20000-25000	40
Total	400

It is clear that the exclusive method ensures continuity of data as much as the upper limit of one class is the lower limit of the next class. In the above example, there are so families whose expenditure is between Rs.0 and Rs.4999.99. A family whose expenditure is Rs.5000 would be included in the class interval 5000-10000. This method is widely used in practice.

b) Inclusive Method:

In this method, the overlapping of the class intervals is avoided. Both the lower and upper limits are included in the class interval. This type of classification may be used for a grouped frequency distribution for discrete variable like members in a family, number of workers in a factory etc., where the variable may take only integral values. It cannot be used with fractional values like age, height, weight etc.

This method may be illustrated as follows:

Class Interval	Frequency
5-9	7
10-14	22
15-19	25
20-24	31
25-29	15

Total	100

Thus to decide whether to use the inclusive method or the exclusive method, it is important to determine whether the variable under observation in a continuous or discrete one. In case of continuous variables, the exclusive method must be used. The inclusive method should be used in case of discrete variable.

c) Open end classes:

A class limit is missing either at the lower end of the first class interval or at the upper end of the last class interval or both are not specified. The necessity of open end classes arises in a number of practical situations, particularly relating to economic and medical data when there are few very high values or few very low values which are far apart from the majority of observations.

The example for the open-end classes as follows:

Salary Range	No. of Workers
Below 2000	7
2000-4000	5
4000-6000	6
6000-8000	4
8000 and above	3
Total	25

Q. How to Convert Inclusive Class Interval to Exclusive Class Interval?

To convert Inclusive Class Interval to Exclusive Class Interval we need to find the average of the gap between the upper and lower limit of two successive classes and then subtract the average from the lower limit of each class and add the average to the upper limit of each class. Let's understand it with an example.

Consider an Inclusive Class Interval given below:

Class Interval	Frequency
10 – 19	12
20 – 29	14
30 – 39	16
40 – 49	11
50 – 59	9

Now to convert this inclusive class interval into exclusive we will find the average of the gap between the upper and lower limit of two successive classes i.e. (10 – 9)/2 = 0.5

Now subtract this 0.5 from the lower limit of each class and add 0.5 to the upper limit of each class. Hence, the exclusive class interval obtained is mentioned below:

Class Interval	Frequency
9.5 – 19.5	12
19.5 – 29.5	14
29.5 – 39.5	16
39.5 – 49.5	11
49.5 – 59.5	9

Cumulative Frequency Table:

Cumulative frequency distribution has a running total of the values. It is constructed by adding the frequency of the first class interval to the frequency of the second class interval. Again add that total to the frequency in the third class interval continuing until the final total appearing opposite to the last class interval will be the total of all frequencies. The cumulative frequency may be downward or upward. A downward cumulating results in a list presenting the number of frequencies "less than" any given amount as revealed by the lower limit of succeeding class interval and the upward cumulative results in a list presenting the number of frequencies "more than" and given amount is revealed by the upper limit of a preceding class interval.

Age Group (in years)	No. of Woman	Less than Cumulative Frequency	More than Cumulative Frequency
15-20	3	3	64
20-25	7	10	61
25-30	15	25	54
30-35	21	46	39
35-40	12	58	18
40-45	6	64	6

(a) Less than cumulative frequency distribution table

End Values Upper Limit	Less than Cumulative Frequency
Less than 20	3
Less than 25	10
Less than 30	25
Less than 35	46
Less than 40	58
Less than 45	64

(b) More than cumulative frequency distribution table

End Values Lower Limit	More than Cumulative Frequency
15 and above	64
20 and above	61
25 and above	54
30 and above	39
35 and above	18
40 and above	6

Relative Frequency Table:

The ratio of the number of times a value of the data occurs in the set of all outcomes to the number of all outcomes gives the value of relative frequency.

Relative Frequency Formula

Relative frequency formula is the formula that is used to find the relative frequency of any given statistical data. We know that relative frequency is defined as the number of times a event occur divided by the ratio of the total event in that case. There are various formulas used to calcu;ate the relative frequency and the formula for relative frequencies are,

Relative Frequency = {Frequency of Given Number(x_i)} / {Sum of frequency of All Quantities (x_1, x_2, x_3, x_4, x_5, x_6…….x_n)}

In other words,

We also calculate the relative frequency by the formula,

Relative Frequency = f/n

where,

- **f** is Frequency of an Observation
- **n** is Total Frequency

How to Calculate Relative Frequency?

To calculate the relative frequency of an object we follow the steps added below,

Step 1: Study the given table and find the frequency of the term of which relative frequency we have to found.

Step 2: Find the total frequency of all the terms from the table.

Step 3: Divide the Frequency of Single Term with the total frequency of all the object to get the required relative frequency.

Relative Frequency Table

The table that contains the relative frequency of all the given elements is called the relative frequency table.

The table added below shows the weight of 30 students of a class along with its relative frequency table and hence it is a Relative Frequency Table.

Weight (in Kg)	Frequency	Relative Frequency
50-55	9	9/30 = 0.3
55-60	7	7/30 = 0.2333
60-65	6	6/30 = 0.2
65-70	2	2/30 = 0.066
70-75	6	6/30 = 0.2

Example : Find the relative frequency of each term from the table. The table added below shows the marks scored by 30 students in a test out of 10.

Marks	Frequency
5	9
6	7
7	6
8	2
9	6

Solution:

The relative frequency of all the terms is added in the table below,

Total Frequency = Total Students = 30

Marks	Frequency	Relative Frequency
5	9	9/30 = 0.3
6	7	7/30 = 0.2333
7	6	6/30 = 0.2
8	2	2/30 = 0.066

Marks	Frequency	Relative Frequency
9	6	6/30 = 0.2

Measures of Central Tendency

Measure of central tendency help you find the middle, or the average of dataset.

Q. Why we need to Measure of central tendency?

It is statistical measure that identifies a single value as representative of entire distribution.

Characteristics of Good Averages

An average is said to be good if it satisfies the following properties-

1. It should be rigidly defined.
2. It should be easy to understand and easy to calculate.
3. It should be based on all items in the data.
4. It should be capable of further mathematical treatment.
5. It should not be affected by extreme (very small or very large) values of the observations.
6. It should be least affected by fluctuation of sampling.
7. It should be capable of further statistical computations or processing.

Various Averages or Measures of Central Tendency or Location-

1. Mean
    a. Arithmetic Mean
    b. Geometric Mean
    c. Harmonic Mean
2. Median
3. Mode

Arithmetic Mean:

Arithmetic mean or simply the mean of a variable is defined as the sum of the observations divided by the number of observations.

I. In Case of Individual Observations (i.e. Where Frequency is not Given)

If the variable x takes the values x1, x2, … xn then arithmetic mean or mean is defined by

$$\boldsymbol{\overline{x} = \frac{x_1+x_2+\cdots+x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}}$$

Example : Calculate the mean for 2, 4, 6, 8, 10.

Solution: $\bar{x} = \frac{x_1+x_2+\cdots+x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n} \quad = \frac{2+4+6+8+10}{5} = \frac{30}{6} = 5.$

II. Case of Discrete Frequency Distribution

If the variable x takes the values x1, x2, … xn occurs f1, f2, … fn times respectively then the mean is defined by

$$\bar{\boldsymbol{x}} = \frac{f_1x_1+f_2x_2+\cdots+f_nx_n}{f_1+f_2\ldots+f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{N}, \text{ where } N = \sum_{i=n}^{n} f_i = \text{total frequency}$$

Example: Given the following frequency distribution, calculate the arithmetic mean

Marks	64	63	62	61	60	59
No. of Students	8	18	12	9	7	6

Solution:

x	f	fx
64	8	512
63	18	1134
62	12	744
61	9	549
60	7	420
59	6	354
Total	60	3713

$$\bar{x} = \frac{f_1x_1+f_2x_2+\cdots+f_nx_n}{f_1+f_2\ldots+f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{N} = \frac{3713}{60} = 61.88.$$

III. In Case of Continuous Frequency Distribution

In continuous frequency distribution the procedure of computing A.M. is the same as in case of discrete frequency distribution. The only difference is that in continuous frequency distribution we find mid-point of each class interval to represent the values of x.

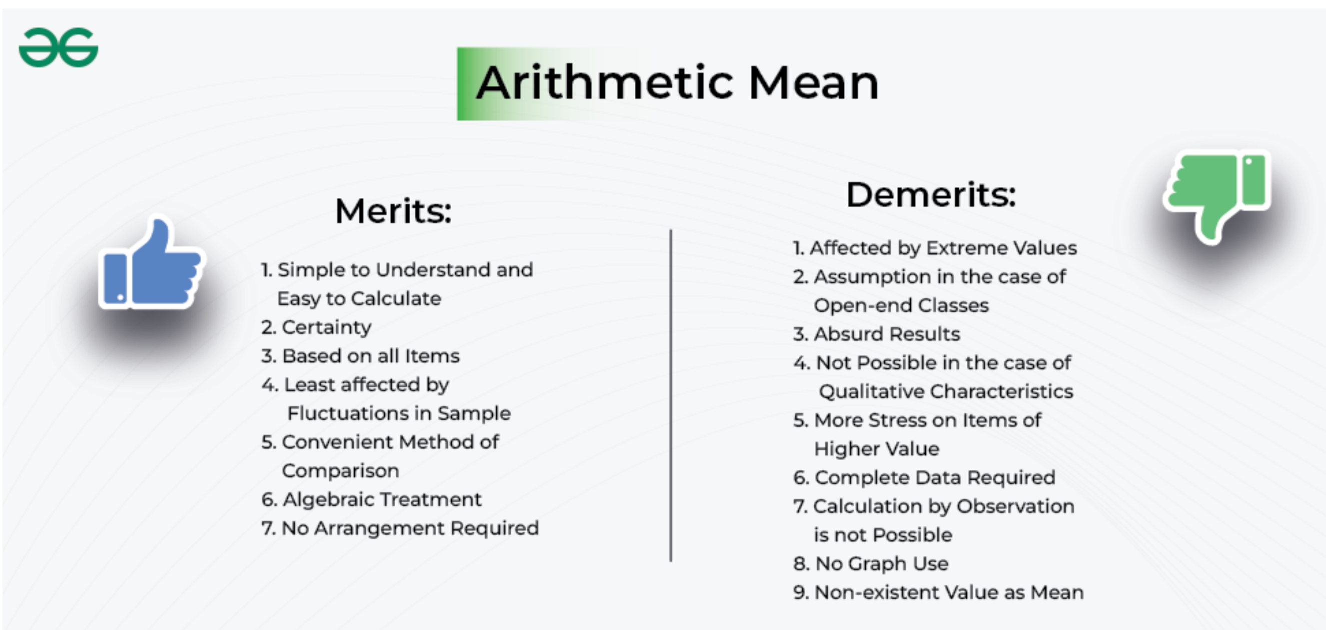
Example: Following is the distribution of persons according to different income groups. Calculate arithmetic mean.

Income Rs.(100)	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of Persons	6	8	10	12	7	4	3

Solution:

Income (C.I.)	No. of Persons (f)	Mid Point (x)	Fx
0-10	6	5	30
10-20	8	15	120
20-30	10	25	250
30-40	12	35	420
40-50	7	45	315
50-60	4	55	220
60-70	3	65	195
Total	50		1550

$$\bar{x} = \frac{f_1x_1+f_2x_2+\cdots+f_nx_n}{f_1+f_2\ldots+f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{N} = \frac{1550}{50} = 31$$



Geometric Mean

The geometric mean of a series containing n observations is the n^{th} root of the product of the values.

I. Case of Individual Observations (i.e. Where Frequency is not Given)

Consider, if x_1, x_2 …. x_n are the observation, then the G.M is defined as:

G.M = $\sqrt[n]{x_1 \times x_2 \times \ldots \times x_n}$ or G.M = $(x_1 \times x_2 \times \ldots \times x_n)^{1/n}$

This can also be written as;

log (G.M) = $\frac{1}{n}\log(x_1 \times x_2 \times \ldots \times x_n) = \frac{1}{n}(logx_1 \times logx_2 \times \ldots \times logx_n) = \frac{\sum \log x_i}{n}$.

Therefore, Geometric Mean,

G.M = Antilog($\frac{\sum \log x_i}{n}$).

Example: Calculate the geometric mean of the following series of monthly income of a batchof families 180, 250, 490, 1400, 1050.

Solution:

x	log x
180	2.2553
250	2.3979
490	2.6902
1400	3.1461
1050	3.0212
Total	13.5107

G.M = Antilog($\frac{\sum \log x_i}{n}$) = Antilog($\frac{13.5107}{5}$) = Anti log 2.7021 = 503.6.

II. In Case of Discrete Frequency Distribution

If the variable x takes the values $x_1, x_2, \ldots x_n$ occurs $f_1, f_2, \ldots f_n$ times respectively then geometric mean is defined by

G.M = $(x_1^{f_1} \times x_2^{f_2} \times \ldots \times x_n^{f_n})^{1/N}$ or G.M = Antilog($\frac{\sum f_i \log x_i}{N}$), where N = $f_1+f_2+\ldots+f_n$

Example : Calculate the average income per head from the data given below. Use geometric mean.

Class of People	Number of Families	Monthly Income per Head (Rs)
Landlords	2	5000
Cultivators	100	400

Landless-Labours	50	200
Money-Leaders	4	3750
Office Assistants	6	3000
Shop Keepers	8	750
Carpenters	6	600
Weavers	10	300

Solution:

Class of People	Annual Income (x)	No. of Families (f)	log x	f log x
Landlords	5000	2	3.6990	7.398
Cultivators	400	100	2.6021	260.210
Landless-Labours	200	50	2.3010	115.050
Money-Leaders	3750	4	3.5740	14.296
Office Assistants	3000	6	3.4771	20.863
Shop Keepers	750	8	2.8751	23.2008
Carpenters	600	6	2.7782	16.669
Weavers	300	10	2.4771	24.771
Total		186		482.257

G.M = Antilog($\frac{\sum f_i \log x_i}{N}$) = Anti log(2.5928) = Rs391.50

III. In Case of Continuous Frequency Distribution

In this case of continuous frequency distribution x is taken to be mid value of each class intervals.

Merits and demerits of Geometric Mean:

Merits:

1. It is rigidly defined.
2. Its is based on all the observations

3. It is capable of further mathematical treatment.
4. It is least affected by fluctuation of sampling.

Demerits:

1. It is difficult to understand and difficult to calculate.
2. It is not defined for negative values of the variable or if the value of the variable is zero.
3. The GM may not be the actual value of the series.

Harmonic Mean

Harmonic mean of a set of observations is defined as the reciprocal of the arithmetic average of the reciprocal of the given values.

I. In Case of Individual Observations (i.e. Where Frequency is not Given)

If x1, x2, x3,…, xn are the individual items up to n terms, then,

$$\textbf{H. M} = \frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n}} = \frac{n}{\sum\frac{1}{x_i}}$$

Example: From the given data calculate H.M. 5, 10, 17, 24, 30.

Solution:

X	$\frac{1}{x}$
5	0.2000
10	0.1000
17	0.0588
24	0.0417
30	0.0333
Total	0.4338

$$\text{H. M} = \frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n}} = \frac{n}{\sum\frac{1}{x_i}} = 5/0.4338 = 11.526$$

II. In Case of Discrete Frequency Distribution

If the variable x takes the values $x_1, x_2, \ldots x_n$ occurs $f_1, f_2, \ldots f_n$ times respectively then geometric mean is defined by

$$\textbf{H. M} = \frac{N}{\frac{f_1}{x_1}+\frac{f_2}{x_2}+\cdots+\frac{fn}{x_n}} = \frac{N}{\sum\frac{f_i}{x_i}}, \text{ where } N = f1+f2+\ldots+fn$$

Example: The marks secured by some students of a class are given below. Calculate the harmonic mean.

Marks	20	21	22	23	24	25
No. of Students	4	2	7	1	3	1

Solution:

Marks (x)	No. of Students (f)	$\frac{1}{x}$	$f.\frac{1}{x}$
20	4	0.0500	0.2000
21	2	0.0476	0.0952
22	7	0.0454	0.3178
23	1	0.0435	0.0435
24	3	0.0417	0.1251
25	1	0.0400	0.0400
Total	18		0.8216

H. M $= \frac{N}{\frac{f_1}{x_1}+\frac{f_2}{x_2}+\cdots+\frac{f_n}{x_n}} = \frac{N}{\sum\frac{f_i}{x_i}}$ = 18/0.8216 = 21.9084

III. In Case of Continuous Frequency Distribution

In this case of continuous frequency distribution x is taken to be mid value of each class intervals.

Merits and Demerits of Harmonic Mean:

Merits:

1. It is rigidly defined.
2. Its is based on all the observations
3. It is capable of further mathematical treatment.
4. It is least affected by fluctuation of sampling.

Demerits:

1. It is difficult to understand and difficult to calculate.
2. The HM may not be the actual value of the series.

Median

Median is defined as the middle term of the given set of data if the data is arranged either in ascending or descending order.

I. In Case of Individual Observations (i.e. Where Frequency is not Given)

Median formula is calculated by two methods,

- Median Formula (when n is Odd)
- Median Formula (when n is Even)

Median Formula (When n is Odd)

If the number of values (n value) in the data set is odd then the formula to calculate the median is,

Median (n = odd number),

$$\text{Median} = \left[\frac{(n+1)}{2}\right]^{th} \text{term}$$

Median Formula (When n is Even)

If the number of values (n value) in the data set is even then the formula to calculate the median is:

Median (n = even number),

$$\text{Median} = \frac{\left[\left(\frac{n}{2}\right)^{th} \text{term} + \left\{\left(\frac{n}{2}\right) + 1\right\}^{th} \text{term}\right]}{2}$$

Example: Find the median of given data set 30, 40, 10, 20, and 50

Solution:

Median of the data 30, 40, 10, 20, and 50 is,

Step 1: Order the given data in ascending order as:

10, 20, 30, 40, 50

Step 2: Check if n (number of terms of data set) is even or odd and find the median of the data with respective 'n' value.

Step 3: Here, n = 5 (odd)

Median = $[(n + 1)/2]^{th}$ term

Median = $[(5 + 1)/2]^{th}$ term = 3^{rd} term

= 30

Thus, the median is 30.

Example: Find median for the following data 5,8,12,30,8, 10, 2, 22.

Solution:

Median of the data 5,8,12,30,8, 10, 2, 22 is,

Step 1: Order the given data in ascending order as:

2,5,8,8,10,12,22,30

Step 2: Check if n (number of terms of data set) is even or odd and find the median of the data with respective 'n' value.

Step 3: Here, n = 8 (even)

Median = $([(n)/2 + (n/2 +1)]^{th}$ term)/2

Median = $([(8)/2\ (8/2 +1)]^{th}$ term)/2 = (4^{th} term + 5^{th} term)/2

= 9

Thus, the median is 9.

II. In Case of Discrete Frequency Distribution

If the variable x takes the values $x_1, x_2, \ldots x_n$ occurs $f_1, f_2, \ldots f_n$ times respectively, then we find median using following steps:

Step 1: Arrange the given distribution in either ascending or descending order.

Step 2: Denote the variables as **X** and frequency as **f**.

Step 3: Determine the cumulative frequency; i.e., **cf.**

Step 4: Calculate the size of (N+1)th/2 item, where $N = f_1+f_2+\ldots + f_n$

Step 5: Then locate the cumulative frequency, which is equal to higher than (N+1)th /2 item and then find the value of x corresponding to this cumulative frequency, which is our required median.

Example :

Calculate the median of the following data:

Income (in ₹)	1000	2000	2500	3000	4500
Number of Workers	6	12	9	14	8

Solution:

Income (in ₹) (X)	Number of Workers (f)	Cumulative Frequency (c.f.)
1000	6	6
2000	12	6 + 12 = 18
2500	**9**	**9 + 18 = 27**
3000	14	14 + 27 = 41
4500	8	41 + 8 = 49
	N = ∑f = 49	

size of (N+1)th/2 item = (49+1)/2=25th item. Now c.f greater than 25 is 27 and its corresponding value of x is 2500.

Hence, Median = 2500

III. In Case of Continuous Frequency Distribution

The steps is given below are followed for the calculation of median in continuous series.

Step 1: Arrange the given data in either descending or ascending order.

Step 2: Determine the cumulative frequency; i.e., **cf.**

Step 3: Calculate the (N/2)th item Where, N = Total of Frequency

Step 4: Now inspect the cumulative frequencies and find out the cf which is either equal to or just greater than the value determined in the previous step.

Step 5: Now, find the class corresponding to the cumulative frequency equal to or just greater than the value determined in the third step. This class is known as the **median class.**

Step 6: Now, apply the following formula for the median:

Median = $l_1 + \frac{\frac{N}{2} - c.f}{f} \times i.$

Where,

l_1 = lower limit of the median class

c.f. = cumulative frequency of the class preceding the median class

f = simple frequency of the median class

i = class size of the median group or class

Note: While calculating the median of a given distribution, we have to assume that every class of the distribution is uniformly distributed in the class interval.

Example:

Class Interval (X)	Frequency (f)
0-5	5
5-10	3
10-15	4
15-20	8
20-25	7
25-30	3

Solution:

Class Interval (X)	Frequency (f)	Cumulative Frequency (c.f.)
0-5	5	5
5-10	3	5 + 3 = 8
10-15	4	8 + 4 = 12 (c.f)
(l_1) 15-20	8 (f)	12 + 8 = 20 Median Class
20-25	7	20 + 7 = 27
25-30	3	27 + 3 = 30
	N = $\sum$f = 30	

Calculate (N/2)th item = (30/2)th item = 15^{th} item.

Hence, the median lies in the class 15-20.

l_1 = 15, f = 8, i = 5, c.f. = 12

Now apply the following formula:

Median = $l_1 + \frac{\frac{N}{2} - c.f}{f} \times i.$

Median = $15 + \frac{\frac{30}{2} - 12}{8} \times 5$

Median= 16.875

Merits of Median

It is one of the easiest and widely used measure of Central Tendency. Some of the merits of Median are as follows:

1. Simple: It is one of the simplest measure of central tendency**.** It is quite easy to compute and simple to understand. It can be easily calculated through inspection in case of some statistical series.

2. Unaffected by Extreme Values: The extreme values do not effect the median of a series. **For example,** median of 20, 25, 30, 35, and 120 is 30; however, its mean is 46. With this, we can see that the median is not affected by the extreme value; i.e., 120, and is thus a better average.

3. Graphical Representation: The Median can be presented graphically through ogive curves, which makes the data more presentable and understandable.

4. Appropriate for Qualitative Data: When dealing with qualitative data, the optimum average to utilize is the median. **For example,** measuring honesty quantitatively is not possible. However, one can easily locate an individual with average honesty through array of a group of persons arranged in ascending or descending order of honesty.

5. Helpful in case of Incomplete or Typical Data: Median is even used in case of incomplete data as median can be calculated even if one knows the number of items and middle item/items of a series. The median is frequently used to express a typical observation. It is influenced mostly by the number of observations rather than their magnitude.

6. Ideal Average: Median has definite and certain value, which means that it is defined rigidly.

Demerits of Median

The demerits of Median are as follows:

1. Arrangement of Data: The given data must be arranged in ascending or descending order in order to compute the Median, which can be a time-consuming task, if the number of items are large.

2. Lack of Representative Character: The median does not represent a measure of such a series in which the values are widely apart.

3. Unpredictable: When the number of items is minimal, the median is unpredictable, which does not depict the true picture of the data.

4. Affected by Fluctuations: The sampling fluctuations have a significant impact on the median. It means that if the class intervals of a series are not uniform, then the median value becomes inappropriate.

5. Lack of further Algebraic Treatment: Median cannot be further algebraically treated. **For example,** just like in case of mean, it is not possible to determine the combined median of two or more groups.

6. Not based on all Observations: As median is a positional average, it is not based on each and every item of a distribution. **For example,** median of 10, 20, 30, 40, and 50 is 30. Now, if we replace the values 10 and 20 with 80 and 90, and replace the values 40 and 50 with 15 and 22 respectively, then it will not have any impact on the median value of the series.

7. Unrealistic assumptions in case of Grouped Distribution: The assumption while determining median in case of grouped distribution is that the median class of the series are uniformly distributed, which in reality rarely happens.

Mode

Mode is the value of the variable which occurs most frequently in a set of observations and around which the other values of the variable have comparatively high frequencies.

Note: There is no mode when all observed value appear in the same number of times in data set.

Note: There is more than one mode when the highest frequency was observed for more than one value of data set.

In both the cases mode can't be used to locate the centre of distribution.

II. In Case of Discrete Frequency Distribution

When the numbers of observations in the series are large then it is difficult to arrange the observations in an array and by inspection one cannot find mode easily. In such situation we form a table of discrete frequency distribution. Then the value of x (observation) having maximum frequency will be taken as mode.

Example: Given the following frequency distribution, calculate the mode

Marks	64	63	62	61	60	59
No. of Students	8	18	12	9	7	6

Solution: Mode = M_0 = 63

III. In Case of Continuous Frequency Distribution

Steps to calculate mode using Observation Method in case of Continuous Series

Step 1: Identify the modal class, which means the class with the highest frequency.

Step 2: The exact value of the mode can be calculated by using the following formula:

$$M_0 = l_1 + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

Where,

M_o = Mode

l_1 = Lower limit of modal class

f_1 = Frequency of modal class

f_0 = Frequency of class preceding the modal class

f_2 = Frequency of the class succeeding the modal class

i = Class interval of the modal class

Example: Find out the mode of the following series.

Class- Interval	Frequency
0-10	7
10-20	10
20-30	15
30-40	8
40-50	9

Solution: By looking at the data, it is evident that the modal class is 20-30 because the frequency of this class is the maximum; i.e., 15.

Class- Interval	Frequency
0-10	7
10-20	10 **(f_0)**
20-30 (l_1)	**15 (f_1)**
30-40	8 **(f_2)**
40-50	9

$$M_0 = l_1 + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

Where, $l_1 = 20$, $f_1 = 15$, $f_2 = 8$, $f_0 = 10$, and $i = 10$. Putting these values in the formula, we get

Mode (M_0) = 24.16

Merits of Mode

Some of the advantages of mode are as follows:

1. Simple and Popular: Mode is a simple measure of central tendency, as one can easily determine the modal value by just glancing at the given series. Its simplicity makes it a very popular measure of central tendency.

2. Graphic Location: One can even locate the mode of a given series through the graph. it can be done with the help of a histogram.

3. No need to know all the Items or Frequencies: To calculate or determine the mode of a given series, one does not require to have knowledge of all the items or frequencies of a series. In simple series, the value with the highest frequency is enough for determining the mode.

4. Least effect of Marginal Values: As compared to the mean, the mode is the measure of central tendency that carries with it the least effect of the marginal values of a series. It is because the mode of a series is determined only through the highest frequencies.

5. Best Representative Value: Mode is that value of a series that occurs most frequently. Therefore, it shows the best representative value of the given series.

6. Possible in the case of Open-end Series: One can determine the mode in open-end distributions without ascertaining the class limits.

7. Useful for both Quantitative and Qualitative Data: One can determine the mode to describe the quantitative as well as qualitative data.

Demerits of Mode

Some of the demerits of the mode are as follows:

1. Uncertain and Vague: Amongst the other measures of central tendency, the mode is vague and uncertain.

2. Difficult: When the frequency of all items in a statistical series is identical, it is difficult for the investigator to identify the modal value of that series.

3. Not capable of further Algebraic Treatment: Unlike mean, one cannot perform further algebraic treatment with mode.

4. Complex Procedure of Grouping: Calculating the mode of a statistical series sometimes also includes the grouping of data, which can be cumbersome, making it difficult to determine the modal value. Also, if the extent of the group changes, it will also change the modal value of the series.

5. Ignores Extreme Marginal Frequencies: The calculation of mode does not consider extreme marginal frequencies. It means that mode is not a representative value of all the values in a given series.

Empirical Relationship between Averages

In a symmetrical distribution the three simple averages mean = median = mode. For a moderately asymmetrical distribution, the relationship between them are brought by Prof. Karl Pearson as mode = 3median- 2mean.

Example 15: If the mean and median of a moderately asymmetrical series are 26.8 and 27.9

respectively, what would be its most probable mode?

Solution: Mode = 3Median – 2Mean

=3*27.9 – 2*26.8 = 30.1

Measures of Dispersion

A measure of central tendency or an average is a single representative value for a set of observations is a frequency distribution. It tells us the centre of the set of data but does not tell us how the set of observations is scattered around this central value. For example, consider the set of scores of two groups of students-

Group A	75	85	95	105	115	125
Group B	10	20	30	70	180	290

The arithmetic mean of each group is 100.
It is observed that in Group A, the values of the variable differ from 100 but the difference is small while in Group B, the values of the variable differ from 100 with high difference.

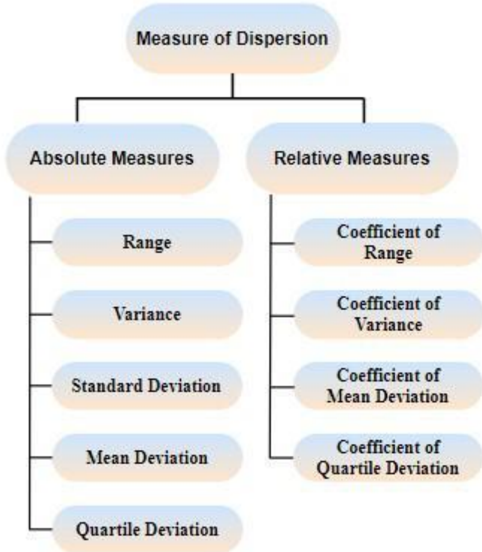
Therefore, while studying a distribution it is equally important to know how the variates are clustered around or scattered away from the point of central tendency. Such measure is called measure of dispersion or measure of variability.

Thus, dispersion is the degree to which the values are dispersed about the central value.

Characteristics of a Good Measure of Dispersion:

An ideal measure of dispersion is expected to possess the following properties

1. It should be rigidly defined.
2. It should be based on all the items.
3. It should not be affected by extreme values.
4. It should lend itself for algebraic manipulation.
5. It should be simple to understand and easy to calculate.



Absolute and Relative Measures:

There are two kinds of measures of dispersion, namely

1. Absolute measure of dispersion
2. Relative measure of dispersion.

Absolute measure of dispersion indicates the amount of variation in a set of values in terms of units of observations. For example, when rainfalls on different days are available in mm, any absolute measure of dispersion gives the variation in rainfall in mm. On the other hand **relative measures** of dispersion are free from the units of measurements of the observations. They are pure numbers. They are used to compare the variation in two or more sets, which are having different units of measurements of observations.

Range:

It is defined as the difference between the largest and the smallest value in the distribution.

Range Formula

Range = Maximum Value (L) – Minimum Value (S).

Co-efficient of Range $= \frac{L-S}{L+S}$

I. In Case of Individual Observations (i.e. Where Frequency is not Given)

This is the simplest possible measure of dispersion and is defined as the difference betweenthe largest and smallest values of the variable.

In symbols, **Range = L – S.**

Where, L = Largest value

S = Smallest value

Example : Find the value of range for the following data 7, 9, 6, 8, 11, 10, 4.

Solution: L=11, S=4

Range = L-S = 11-4 = 7

Coefficient of range = $\frac{L-S}{L+S} = \frac{7}{15}$ = 0.4667

II. In Case of Discrete Frequency Distribution

It is defined as the difference betweenthe largest and smallest values of the variable i.e, L-S.

Example: Weekly wages of labours are given below. Find the value of range.

Weekly Wages (Rs.)	100	200	300	400	500
No. of Weeks	5	8	21	12	6

Solution: L= 500, S= 100

Range = L-S = 500 – 100 = 400.

Coefficient of range = $\frac{L-S}{L+S} = \frac{400}{600}$ = 0.6667

III. In Case of Continuous Frequency Distribution

In continuous series,

Range = L-S

L = Upper boundary of the highest class
S = Lower boundary of the lowest class

Example: Calculate range from the following distribution

Size	60-63	63-66	66-69	69-72	72-75
Number	5	18	42	27	8

Solution: L = Upper boundary of the highest class = 75

S = Lower boundary of the lowest class = 60

Range = L – S = 75 – 60 = 15

Merits of Range

1. It is easy to understand and calculate.
2. It provides a quick measure of variability.
3. Range provides an overview of the data at once.

Demerits of Range

1. Range is not based on all of the observations. The range of distribution remains the same if every item is changed except for the smallest and largest item.
2. Fluctuations in sampling have a big impact on range. Its value differs widely between samples.
3. It does not provide any insight into the pattern of distribution. It is possible for two distributions to have the same range but different patterns of distribution.

Quartile Deviation

Quartile Deviation or Semi-Interquartile Range is the half of difference between the Upper Quartile (Q_3) and the Lower Quartile (Q_1). In simple terms, QD is the half of inter-quartile range. Hence, the formula for determining Quartile Deviation is as follows:

Quartile Deviation $= \frac{Q_3 - Q_1}{2}$

Where,

Q_3 = Upper Quartile (Size of 3[(N+1)/4] item)

Q_1 = Lower Quartile (Size of [(N+1)/4] item)

What is Coefficient of Quartile Deviation?

Coefficient of Quartile Deviation $= \frac{Q_3 - Q_1}{Q_3 + Q_1}$

Where,

Q_3 = Upper Quartile (Size of 3[(N+1)/4] item)

Q_1 = Lower Quartile (Size of [(N+1)/4] item)

Calculation of Quartile Deviation in Different Series

I. In Case of Individual Observations (i.e. Where Frequency is not Given)

Example:

With the help of the data given below, find the quartile deviation, and coefficient of quartile deviation.

150	100	268	280	195	140	200

Solution:

S.No.	Items arranged in ascending order
1	100
2	140
3	150
4	195
5	200
6	268
7	280
N = 7	

Q_1 = Size of [(N+1)/4] item = Size of [(7+1)/4] item = 2^{nd} Item

$Q_1 = 140$

Q_3 = Size of 3[(N+1)/4] item = Size of 3[(7+1)/4] item = 6^{th} Item

$Q_3 = 268$

Quartile Deviation $= \frac{Q_3 - Q_1}{2} = \frac{268-140}{2} = 64$

Coefficient of Quartile Deviation $= \frac{268-140}{268+140} = \frac{128}{408} = 0.31$

Mean Deviation:

It is the arithmetic mean of the difference between the values and their mean.

Coefficient of mean deviation:

Coefficient of mean deviation = Mean deviation /Mean or Median or Mode.

I. In Case of Individual Observations (i.e. Where Frequency is not Given)

Mean Deviation (M.D) = $\sum|X - \bar{X}| / N$

Where,

$\sum$ – Summation

x – Observation

$\bar{X}$ – Mean or Meadian or Mode

N – Number of observation

Example: Calculate the mean deviation from the median and the co-efficient of mean deviation from the following data: Marks of the students: 88, 14, 78, 69, 44, 54, 18, 79, 40.

Solution:

Arrange the data in ascending order: 14, 18, 40, 44, 54, 69, 78, 79, 88.

Median = Value of the $(N + 1)^{th} / 2$ term

= Value of the $(9 + 1)^{th} / 2$ term = 54

Calculation of mean deviation:

X	$\lvert X – M \rvert$
14	40
18	36
40	14
44	10
54	0
69	15
78	24
79	25
88	34
N = 9	$\sum\lvert X–M\rvert=198$

M.D. = $\sum|X – M| / N$

= 198/9

= 22

Co-efficient of Mean Deviation from Median = M.D./M

= 22/54

= 0.4074

II. In Case of Discrete Frequency Distribution

Mean Deviation (M.D) = $\sum f|X - M| / \sum f$

Where,

$\sum$ – Summation

x – Observation

M – Median or Mean or Mode

N – Frequency of observations

Example:

Calculate the mean deviation from mean and median for the following distribution.

Values (X)	Frequency (f)
5	5
6	11
7	15
8	14

Solution:

Mean Deviation from the Mean

Calculation of Mean Deviation from the Mean

Values (X)	Frequency (f)	fX	$\lvert D\rvert = \lvert X-\bar{X}\rvert$	f$\lvert D\rvert$
5	5	25	1.85	9.25
6	11	66	0.85	9.35
7	15	105	0.15	2.25
8	14	112	1.15	16.1
	Σf = 45	**ΣfX = 308**		**Σf$\lvert$D$\rvert$ = 36.95**

Mean(M) = 6.85

Mean Deviation (M.D) = $\sum f|X - M| / \sum f$ =36.5/45 = 0.82

Mean Deviation from Mean = 0.82

III. In Case of Continuous Frequency Distribution

Example:

Calculate the mean deviation from the mean and median for the following data:

Class Interval (X)	Frequency (f)
0-10	2
10-20	3
20-30	4
30-40	5

Solution:

Mean Deviation from the Mean

Calculation of Mean Deviation from Mean

Class Interval (X)	Frequency (f)	Mid-point (m)	fm	$\lvert D\rvert = \lvert m-\bar{X}\rvert$	$f\lvert D\rvert$
0-4	2	2	4	7.42	14.84
4-8	3	6	18	3.42	10.26
8-12	4	10	40	0.58	2.32
12-16	5	14	70	4.58	22.9
	Σf = 14		**Σfm = 132**		**Σf|D| = 50.32**

Mean = 9.42

Mean Deviation from Mean = 3.59

Mean Deviation from the Median

Calculation of Mean Deviation from Median

Class Interval (X)	Frequency (f)	Cumulative Frequency (cf)	Mid-point (m)	|D|= |m-Me|	f|D|
0-4	2	2	2	8	16
4-8	3	5	6	4	12
8-12	4	9	10	0	0
12-16	5	14	14	4	20
	Σf = 14				**Σf|D| = 48**

Median = 10

Mean Deviation from Median = 3.42

Merits and Demerits of Mean Deviation:

Merits:

1. It is simple to understand and easy to calculate.
2. It is rigidly defined.
3. It is based on all items of the series.
4. It is not much affected by the fluctuations of sampling.
5. It is less affected by the extreme items.
6. It is flexible, because it can be calculated from any average.
7. It is better measure of comparison.

Demerits:

1. It is not a very accurate measure of dispersion.
2. It is not suitable for further mathematical calculation.
3. It is rarely used. It is not as popular as standard deviation.
4. Algebraic positive and negative signs are ignored. It is mathematically unsound and illogical.

Standard Deviation:

"The Standard Deviation is the square root of the arithmetic mean of the squares of all deviations. Deviations being measured from arithmetic mean of the items."

Coefficient of Standard Deviation is a relative measure of Standard Deviation and is determined by dividing Standard Deviation by the Mean of the given data set. It is also known as the **Standard Coefficient of Dispersion.**

Coefficient of Standard Deviation = $\frac{\sigma}{\bar{x}}$

I. In Case of Individual Observations (i.e. Where Frequency is not Given)

Actual Mean Method:

In actual mean method, standard deviation is calculated by taking deviations from the actual mean(arithmetic mean). The steps taken to determine standard deviation through the actual mean method are as follows:

Step 1: Determine the arithmetic mean $\bar{x}$ of the given observation.

Step 2: Now, calculate the deviation of each item of the given series from the mean calculated in the first step; i.e., calculate $(x - \bar{x})$.

Step 3: Apply the following formula:

$$\sigma = \sqrt{\frac{\sum(x - \overline{x})^2}{N}}$$

Where,

σ = Standard Deviation

$\sum(x - \overline{x})^2$ = Sum total of the squares of deviations from the arithmetic mean

N = Number of pairs of observations

Example:

Calculate the standard deviation and coefficient of standard deviation from the following data:

5	8	3	6	7	14	2	5

Solution:

Standard Deviation (Actual Mean Method)

X	$x = X - \bar{X}$	**x^2**
5	5 - 6 = -1	1
8	8 - 6 = 2	4
3	3 - 6 = -3	9
6	6 - 6 = 0	0
7	7 - 6 = 1	1
12	12 - 6 = 6	36
2	2 - 6 = -4	16
5	5 - 6 = -1	1
∑X = 48		$\sum x^2 = 68$

Arithmetic Mean ($\overline{x}$) =6

Therefore, Standard deviation is

$$\sigma = \sqrt{\frac{\sum(x-\bar{x})^2}{N}} = \sqrt{\frac{68}{48}} = \sqrt{8.5} = 2.9$$

Coefficient of Standard Deviation = $\frac{2.9}{6}$ = 0.483.

II. In Case of Discrete Frequency Distribution

Step 1: Determine the actual mean $\bar{x}$ of the given observation, using the formula, $\left(\frac{\sum fx}{N}\right)$

Step 2: Now, calculate the deviation of each item of the given series from the mean calculated in the first step; i.e., calculate $(x - \bar{x})$. Denote the deviations with x.

Step 3: After finding out the deviation, square these deviations and multiply them with their respective frequencies (f), and determine its total; i.e., $\sum f(x - \bar{x})^2$.

Step 4: Apply the following formula:

$$\sigma = \sqrt{\frac{\sum f(x-\bar{x})^2}{N}}$$

Where,

σ = Standard Deviation

$\sum f(x - \bar{x})^2$= Sum total of the squared deviations multiplied by frequency

N = Sum of frequency.

Example:

Calculate the Standard Deviation for the following data by using the Actual Mean Method.

Size (X)	5	10	15	20	25	30
Frequency (f)	6	7	3	2	1	1

Solution:

Size (X)	Frequency (f)	fX	$x = X - \bar{X}$	x^2	fx^2
5	6	30	5 - 12 = -7	49	294
10	7	70	10 - 12 = -2	4	28
15	3	45	15 - 12 = 3	9	27
20	2	40	20 - 12 = 8	64	128
25	1	25	25 - 12 = 13	169	169
30	1	30	30 - 12 = 18	324	324
	N = ∑f = 20	**∑fX = 240**			**$\sum fx^2 = 970$**

Arithmetic Mean ($\bar{x}$) =12

Standard Deviation =

$$\sigma = \sqrt{\frac{\sum f(x-\bar{x})^2}{N}} = \sqrt{\frac{970}{20}} = \sqrt{48.5} = 6.9$$

Coefficient of Standard Deviation = $\frac{6.9}{12}$ = 0.575

III. In Case of Continuous Frequency Distribution

In case of continuous frequency distribution, the values of mid-values of each of the n class intervals and use the same formula as of discrete distribution.

Example:

Calculate the standard deviation of the following series by Direct Method.

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	2	7	10	5	3

Solution:

Class Interval	Mid-Value (m)	Frequency (f)	fm	$x = m - \bar{X}$ $\bar{X} = 32.5$	x^2	fx^2
0 - 10	5	2	10	-20	400	800
10 - 20	15	7	105	-10	100	700
20 - 30	25	10	250	0	0	0
30 - 40	35	5	175	10	100	500
40 - 50	45	3	135	20	400	1200
		N = 27	**$\sum fm = 675$**			**$\sum fx^2 = 3200$**

Standard Deviation = 10.89

Merits of Standard Deviation

The merits of Standard Deviation are:

1. Based on all Values: Every item of a series is considered in Standard Deviation. Therefore, a change in even one value of the series affects the value of standard deviation.

2. Algebraic Treatment: Further algebraic treatment is possible in case of standard deviation. **For example,** if one knows the standard deviation of different groups, their combined standard deviation can be easily determined.

3. Better Mathematical Process: The drawback of ignoring signs of deviations(as done in mean deviation) is removed in standard deviation through squaring of deviations.

4. Rigidly Defined: By far, standard deviation is the most essential and widely used measure of dispersion. It is a definite measure of dispersion; hence, is rigidly defined.

5. Less effect of Fluctuations in Sampling: If various independent samples are drawn from the same population, then we can see that the standard deviation of the distribution is least affected from one sample to another as compared to the effect caused by other measures of dispersion.

Demerits of Standard Deviation

The demerits of Standard Deviation are:

1. Difficult to Compute: As compared to other measures of dispersion, it is difficult to compute standard deviation.

2. Depend upon Units of Measurement: Standard Deviation of a series depends upon the units of measurement of the observations. Therefore, one cannot use standard deviation for comparing the dispersion of the distributions that are expressed in different units.

3. More Stress on Extreme Values: Standard Deviation gives more weightage to the extreme values and less weightage to the values which are nearer to mean.

Variance:

The square of the standard deviation is known as variance. Therefore, the variance is defined as the arithmetic mean of the squares of deviations of the observations from the arithmetic mean.

Variance = (standard deviation)^2 = σ^2

Coefficient of variance $= \frac{\sigma}{\bar{x}} \times 100$